

REQUIREMENT REPORT

Complaint Management Portal

1. Project Title

Complaint Management Portal

2. Introduction

The Complaint Management Portal is a web-based software system designed to provide an efficient platform for users to register, track, manage, and resolve complaints digitally.

The system allows users to submit complaints related to different departments or services. Administrators and department staff can view complaints, assign them to responsible personnel, update complaint status, communicate with users, and monitor the overall resolution process.

The main objective of the system is to replace manual complaint handling with a centralized, transparent, and efficient digital system.

3. Problem Statement

In a traditional complaint management system, complaints are often submitted through physical forms, phone calls, emails, or verbal communication. This can result in:

- Delay in complaint registration
- Loss or duplication of complaints
- Difficulty in tracking complaint status
- Lack of transparency
- Poor communication between users and departments
- Difficulty in maintaining complaint records
- No proper monitoring of unresolved complaints
- Difficulty in generating reports and statistics

Therefore, a centralized Complaint Management Portal is required to automate the complaint registration and resolution process.

4. Objectives

The major objectives of the Complaint Management Portal are:

1. To provide an online platform for registering complaints.
 2. To allow users to track complaint status.
 3. To provide administrators with centralized complaint management.
 4. To assign complaints to appropriate departments/staff.
 5. To maintain complete complaint history.
 6. To reduce complaint resolution time.
 7. To provide notifications regarding complaint updates.
 8. To improve transparency between users and authorities.
 9. To generate complaint-related reports.
 10. To provide feedback after complaint resolution.
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5. Stakeholders

Stakeholder	Role
User/Complainant	Registers and tracks complaints
Administrator	Manages the complete system
Department Staff	Handles assigned complaints
Department Head	Monitors departmental complaints
System	Stores, processes and manages complaint information

6. User Requirements

6.1 User Registration

The system shall allow a new user to create an account by providing:

- Name
- Email
- Mobile Number
- Password
- Address/Location where required

6.2 User Login

Registered users shall be able to securely log into the system using their credentials.

6.3 Register Complaint

Users shall be able to create a complaint by entering:

- Complaint title
- Complaint description
- Complaint category
- Department
- Location
- Priority
- Supporting image/document, if applicable

6.4 Complaint Tracking

Users shall be able to view:

- Complaint ID
- Complaint date

- Complaint category
- Assigned department
- Current status
- Resolution details

6.5 Complaint History

Users shall be able to view their previously submitted complaints.

6.6 Notifications

The system shall notify users whenever there is an important update in their complaint.

6.7 Feedback

After complaint resolution, users shall be able to provide feedback and rating.

7. Administrator Requirements

The administrator shall be able to:

1. Login securely.
2. Manage registered users.
3. View all complaints.
4. Search and filter complaints.
5. Categorize complaints.
6. Assign complaints to departments.
7. Assign complaints to staff members.
8. Change complaint status.
9. Monitor pending complaints.
10. Monitor resolved complaints.
11. Handle escalated complaints.
12. View complaint statistics.
13. Generate reports.
14. Manage departments and staff.
15. Manage system notifications.

8. Department Staff Requirements

Department staff shall be able to:

- Login to the portal.
- View assigned complaints.
- View complaint details.
- Accept assigned complaints.
- Update complaint status.
- Add comments or resolution details.
- Upload supporting documents/images.
- Mark complaints as resolved.
- View complaint history.

9. Functional Requirements

ID	Functional Requirement	Description	Priority
FR-01	User Registration	User can create an account	High
FR-02	User Login	User can securely login	High
FR-03	Admin Login	Admin can access administrative functions	High
FR-04	Complaint Registration	User can submit a complaint	High
FR-05	Complaint ID Generation	System generates a unique ID	High
FR-06	Complaint Categorization	Complaints can be categorized	Medium
FR-07	Department Assignment	Complaint can be assigned to department	High
FR-08	Staff Assignment	Complaint can be assigned to staff	High
FR-09	Complaint Tracking	User can track complaint status	High
FR-10	Status Update	Staff can update complaint status	High
FR-11	Complaint History	System maintains complaint history	High
FR-12	Search Complaint	Users/admin can search complaints	Medium
FR-13	Filter Complaint	Complaints can be filtered	Medium
FR-14	Notification	System sends complaint updates	Medium
FR-15	Feedback	User can provide feedback	Medium
FR-16	Escalation	Unresolved complaints can be escalated	Medium

ID	Functional Requirement	Description	Priority
FR-17	Report Generation	Admin can generate reports	Medium
FR-18	User Management	Admin can manage users	High
FR-19	Department Management	Admin can manage departments	Medium

10. Non-Functional Requirements

ID	Requirement	Description
NFR-01	Performance	System should respond quickly to user requests
NFR-02	Security	User data and passwords must be protected
NFR-03	Availability	Portal should be available whenever required
NFR-04	Usability	Interface should be simple and user-friendly
NFR-05	Scalability	System should support increasing users and complaints
NFR-06	Reliability	Complaint information should not be lost
NFR-07	Maintainability	System should be easy to maintain and update
NFR-08	Compatibility	System should work on modern web browsers
NFR-09	Privacy	User information should be accessible only to authorized users

11. Data Requirements

The system shall maintain the following major data entities.

11.1 User

- User ID
- Name
- Email
- Mobile Number
- Password
- Address
- Registration Date

11.2 Complaint

- Complaint ID

- User ID
- Complaint Title
- Description
- Category
- Department
- Location
- Priority
- Status
- Created Date
- Updated Date
- Resolution Details

11.3 Department

- Department ID
- Department Name
- Department Description
- Department Contact

11.4 Staff

- Staff ID
- Name
- Email
- Department
- Contact
- Role

11.5 Feedback

- Feedback ID
- Complaint ID
- User ID
- Rating
- Comment
- Feedback Date

11.6 Notification

- Notification ID
 - User ID
 - Complaint ID
 - Message
 - Notification Date
 - Read/Unread Status
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12. Complaint Status Requirements

The complaint shall follow a defined workflow:

Submitted → Under Review → Assigned → In Progress → Resolved → Closed

If the complaint is not resolved within the defined time, it may be:

Escalated → Higher Authority Review

The system shall maintain the history of every status change.

13. Use Case Requirements

Use Case ID	Use Case	Actor
UC-01	Register Account	User
UC-02	Login	User/Admin/Staff
UC-03	Submit Complaint	User
UC-04	View Complaint	User/Admin/Staff
UC-05	Track Complaint	User
UC-06	Assign Complaint	Admin
UC-07	Update Complaint	Staff
UC-08	Resolve Complaint	Staff
UC-09	Escalate Complaint	Admin
UC-10	Send Notification	System
UC-11	Provide Feedback	User
UC-12	Manage Users	Admin
UC-13	Manage Departments	Admin

Use Case ID	Use Case	Actor
UC-14	Generate Reports	Admin

14. External Interface Requirements

14.1 User Interface

The portal shall provide:

- Login page
- Registration page
- User dashboard
- Complaint submission form
- Complaint tracking page
- Complaint history
- Admin dashboard
- Staff dashboard
- Notification section
- Feedback section

14.2 Hardware Interface

The system does not require specialized hardware. It can operate on:

- Desktop computers
- Laptops
- Tablets
- Smartphones

with internet access.

14.3 Software Interface

The system may use:

- Web Browser
 - Frontend application
 - Backend/API server
 - Database server
 - Notification service
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15. Security Requirements

The system shall provide:

1. Secure user authentication.
 2. Password protection.
 3. Role-based access control.
 4. Authorization for administrative operations.
 5. Protection against unauthorized access.
 6. Secure storage of user information.
 7. Input validation.
 8. Secure session/token management.
 9. Complaint data privacy.
 10. Proper logout functionality.
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16. System Constraints

The system may have the following constraints:

- Internet connection is required.
 - Users must have valid login credentials for protected features.
 - Only authorized staff can modify assigned complaints.
 - Admin privileges shall be restricted.
 - System performance depends on server and database availability.
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17. Assumptions

The system assumes that:

- Users provide correct information.
 - Departments have authorized staff members.
 - Administrators maintain department and staff information.
 - Users have access to the internet.
 - The database server is available.
 - Users regularly check notifications or receive enabled notifications.
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18. Acceptance Requirements

The Complaint Management Portal shall be considered acceptable when:

- Users can register successfully.
 - Users can login successfully.
 - Users can submit complaints.
 - A unique Complaint ID is generated.
 - Admin can view submitted complaints.
 - Admin can assign complaints to departments/staff.
 - Staff can update complaint status.
 - Users can track complaint progress.
 - Resolved complaints can be closed.
 - Users can provide feedback.
 - Complaint records are stored correctly.
 - Unauthorized users cannot access restricted functions.
 - Admin can view complaint reports.
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19. Future Requirements

Future versions of the system may include:

- AI-based complaint categorization
- AI-based priority prediction
- Automatic department assignment
- Chatbot support
- Duplicate complaint detection
- Sentiment analysis
- Mobile application
- SMS notifications
- Email notifications
- Multilingual support
- Voice-based complaint registration
- Predictive complaint resolution time
- Advanced analytics dashboard
- Automatic escalation

- AI-based complaint recommendation and routing
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20. Conclusion

The Complaint Management Portal will provide a centralized digital platform for managing complaints from registration to resolution. It will reduce manual work, improve transparency, provide complaint tracking, and help administrators and departments manage complaints efficiently.

The defined functional and non-functional requirements provide a foundation for designing, developing, testing, and maintaining the Complaint Management Portal.